

1. Key Decisions of the Solitaire Project

TOPIC	DECISION
Object-oriented programming language (if not using OOP, then justify the use of the language)	Python
GUI library (strongly encouraged)	Tkinter
IDE (Integrated Development Environment)	Visual Studio Code
xUnit framework (e.g. JUnit for Java)	unittest
Programming style guide (must read it carefully)	Google Python Style Guide
Project hosting site	Github.com
Other decisions if applicable	n/a

2. Unit Testing

Source Code:

```
Sprint 0.py
C: > Users > jmlam > OneDrive > Desktop > Sprint 0.py > ...
1  import unittest
2
3
4  def add(a, b):
5      return a + b
6
7
8  def subtract(a, b):
9      return a - b
10
11
12 class TestCalculator(unittest.TestCase):
13
14     def test_add(self):
15         self.assertEqual(add(5, 3), 8)
16
17     def test_subtract(self):
18         self.assertEqual(subtract(5, 3), 2)
19
20
21 if __name__ == "__main__":
22     unittest.main()
23
24
```

Output:

```
[Running] python -u "c:\Users\jmlam\OneDrive\Desktop\Sprint 0.py"
..
-----
Ran 2 tests in 0.000s

OK

[Done] exited with code=0 in 0.364 seconds
```

3. GUI Programming

Source Code:

```
1  import tkinter as tk
2
3  # Create the main window
4  window = tk.Tk()
5  window.title("Simple GUI")
6  window.geometry("400x300")
7
8  # Text
9  title = tk.Label(window, text="Solitaire GUI Practice", font=("Arial", 18))
10 title.pack(pady=10)
11
12 # Canvas for lines and text
13 canvas = tk.Canvas(window, width=350, height=80)
14 canvas.pack()
15
16 # Lines
17 canvas.create_line(50, 20, 300, 20, width=2)
18 canvas.create_line(50, 60, 300, 60, width=2)
19
20 # Text on the canvas
21 canvas.create_text(175, 40, text="Card Area")
22
23 # Check box
24 sound_var = tk.BooleanVar()
25 sound_checkbox = tk.Checkbutton(
26     window,
27     text="Enable Sound",
28     variable=sound_var
29 )
30 sound_checkbox.pack(pady=5)
31
32 # Radio buttons
33 difficulty = tk.StringVar(value="Easy")
34
35 easy_radio = tk.Radiobutton(
36     window,
37     text="Easy",
38     variable=difficulty,
39     value="Easy"
40 )
41 easy_radio.pack()
42
43 hard_radio = tk.Radiobutton(
44     window,
45     text="Hard",
46     variable=difficulty,
47     value="Hard"
48 )
49 hard_radio.pack()
50
51 # Start the GUI
52 window.mainloop()
```

Output:

